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Kelas: 1A - TTE

No. :

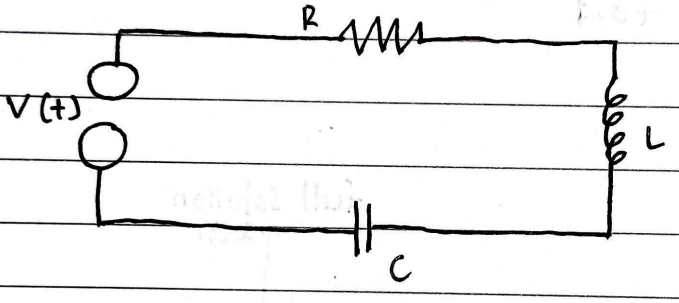
Tugas Fisika Terapan (Getaran Listrik)

1. The charge $Q(t)$ on the capasitor in the electric circuit shown satisfies the differential equation

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{1}{C}Q = V(t)$$

Where { $L = 0,5$ hendrys is the coil's inductance
$R = 6.0$ ohms is the resistor's resistance
$C = 0,02$ farads is the capasitor's capasitance and

$V(t)$ is the applied voltage



(i) is the circuit oscillatory

(ii) if $(v)(t) = 24 \sin(10t)$ volt and $Q(0) = 0, Q'(0)$ find $Q(t)$

(iii) sketch the transient solution the steady state solution and the full solution $Q(t)$.

Answer:

(i). $R_C = \sqrt{\frac{4L}{C}} = \sqrt{\frac{4.(0,5)}{0,02}} = \sqrt{100} = 10\Omega$

$R \angle R_C$
$6 \angle 10$

(ii) $24^6\angle 0^\circ = L\frac{d^2Q}{dt} + R\frac{dQ}{dt} + \frac{1}{C}Q$

$Z = R + j(X_L - X_C)$
$= 6 + j(\omega L - \frac{1}{\omega C})$
$= 6 + j\left(10.0,5 - \frac{1}{10.0,02}\right)$
$= 6 + j(0)$
$= 6$

$i_t = \frac{24\angle 0^\circ}{6\angle 0^\circ}$
$= 4\angle 0^\circ$
$\therefore 4 \sin \omega t = 4 \sin 10t$

180 mm x 257 mm

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No. :

$$Q(t) = \int_0^t 4 \sin 10t$$

$$= 4 \int_0^t \sin 10t$$

$$= -\frac{4}{10} \left(\cos 10t \right]_0^t + C$$

$$= -0,4 \cos 10t + C$$

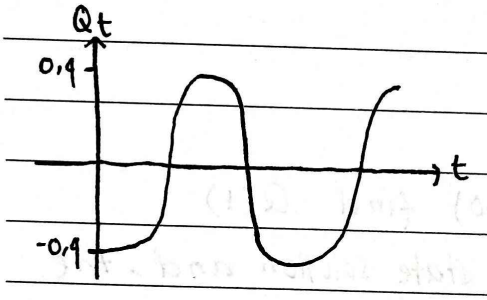
$$-C = -0,4 \cos 0$$

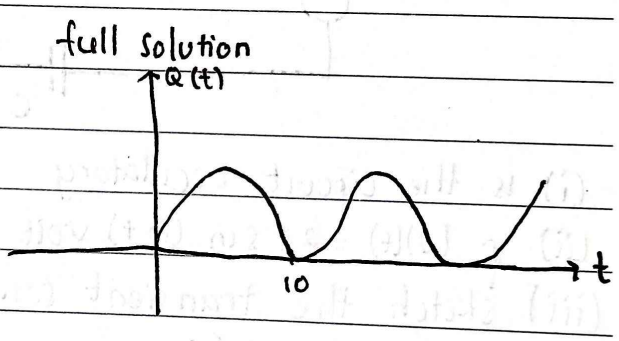
$$= -0,4$$

$$C = 0,4$$

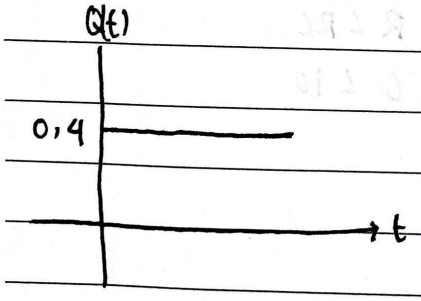
maka $Q(t) = -0,4 \cos 10t + 0,4$

(iii) Transient Solution





Stedy state Solution



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